homework04

P143-习题 8.8　(A 类)　1-1, 1-3; 2; 9-1; 12; 15; 19
P185-习题 9.1　(A 类)　7-1, 7-4, 7-5;

1-1、极值 $z = x^3 + 3xy^2 - 15x - 12y$

解: 依题意, 由

$$z_x = 3x^2 + 3y^2 - 15 = 0, \quad z_y = 6xy - 12 = 0$$

$$\Rightarrow (x, y) = \pm(1, 2) \ \pm (2, 1) \quad z_{xx} = 6x, \ z_{xy} = 6y, \ z_{yy} = 6x,$$

$$\Rightarrow z_{xx}z_{yy} - z_{xy}^2 = 36(x^2 - y^2)\Big|_{\pm(2,1)} > 0 \Rightarrow z_{xx}z_{yy} - z_{xy}^2 = 36(x^2 - y^2)\Big|_{\pm(1,2)} < 0$$

$$\Rightarrow z\text{在 } (2, 1) \text{ 取极小值 } -28, \text{ 在 } (-2, -1) \text{ 取极大值 } 28$$

1-3、极值 $z = 6x^2 + 9y^2 + 4xy - 32x - 64y + 256$

解: 依题意, 由

$$z_x = 12x + 4y - 32 = 0, \quad z_y = 18y + 4x - 64 = 0 \Rightarrow (x, y) = (\frac{8}{5}, \frac{16}{5})$$

$$\Rightarrow z_{xx} = 12, \quad z_{yy} = 18, \quad z_{xy} = 4 \Rightarrow z_{xx}z_{yy} - z_{xy}^2 > 0$$

$$\Rightarrow z\text{在 } (\frac{8}{5}, \frac{16}{5}) \text{ 取极小值 } 128$$

2、求由方程 $x^2+y^2+z^2-2x+4y-6z-11=0$ 所确定的隐函数的极值.

解: 依题意

$$\text{求偏导} \Rightarrow 2x+2zz_x-2-6z_x=0, \quad 2y+2zz_y+4-6z_y=0$$

$$\text{由} z_x=z_y=0 \Rightarrow (x,y)=(1,-2) \Rightarrow z=-2 \text{ 或 } 8$$

$$\text{求二阶偏导} \Rightarrow 2+2z_x^2+2zz_{xx}-6z_{xx}=0, \quad 2z_yz_x+2zz_{xy}-6z_{xy}=0,$$

$$2+2z_y^2+2zz_{yy}-6z_{yy}=0$$

$$\Rightarrow \text{在 } (x,y)=(1,-2) \text{ 处} \quad z_{xx}=\frac{1}{3-z}, \quad z_{xy}=0, \quad z_{yy}=\frac{1}{3-z}$$

$$z=-2\text{时}, \ z_{xx}z_{yy}<z_{xy}^2, \ z_{xx}=\frac{1}{3} \Rightarrow z \text{ 在 } (1,-2,-2) \text{ 处取极小值 } -2$$

$$z=8\text{时}, \ z_{xx}z_{yy}<z_{xy}^2, \ z_{xx}=\frac{1}{-5} \Rightarrow z \text{ 在 } (1,-2,8) \text{ 处取极大值 } 8$$

9-1、求条件极值 $z=x^2+y^2$ 约束条件: $\frac{x}{a}+\frac{y}{b}=1,(a,b>0)$

解: 依题意, 记 $L=x^2+y^2+\lambda(\frac{x}{a}+\frac{y}{b}-1)$

$$L_x=2x+\frac{\lambda}{a}=0,\quad L_y=2y+\frac{\lambda}{b}=0,\quad \frac{x}{a}+\frac{y}{b}=1$$

$$\Rightarrow ax=by\Rightarrow x=\frac{ab^2}{a^2+b^2},\quad y=\frac{a^2b}{a^2+b^2},\quad \lambda=\frac{-2x}{a}=\frac{-2a^2b^2}{a^2+b^2}$$

$$\Rightarrow \text{由几何意义知 } z \text{ 在 } (\frac{ab^2}{a^2+b^2},\frac{a^2b}{a^2+b^2}) \text{ 取极小值为 } \frac{a^2b^2}{a^2+b^2}$$

12、求 $f(x,y)=x^2+y^2-12x+16y$ 在 $D=\{(x,y)|x^2+y^2\le 25\}$ 上的最值.

解: 依题意, 取 $L=f(x,y)+\lambda(x^2+y^2-25)$

$$\text{由}\quad f_x=2x-12=0, f_y=2y+16=0\Rightarrow (x,y)=(6,-8)\text{舍}$$

$$\text{由 } L_x=2(1+\lambda)x-12=0,\ L_y=2(1+\lambda)y+16=0,\ L_\lambda=x^2+y^2-25=0$$

$$\Rightarrow \lambda+1\neq 0\Rightarrow 4x+3y=0\Rightarrow (x,y)=\pm(3,-4),\ \lambda=\frac{6}{x}-1$$

$$f(3,-4)=-75,\quad f(-3,4)=125$$

$$\Rightarrow f_{\min}=f(3,-4)=-75,\quad f_{\max}=f(-3,4)=125$$

15、求 $f(x,y,z)=x-y+z$ 在区域 $\Omega:\frac{(x-1)^2}{4}+\frac{(y+1)^2}{4}+(z-2)^2\le 1$ 上的最值.

解: 依题意, 取 $L=f(x,y,z)+\lambda\big(\dfrac{(x-1)^2}{4}+\dfrac{(y+1)^2}{4}+(z-2)^2-1\big)$

$$\text{由}\quad f_x=1\neq 0\Rightarrow D\text{ 内无驻点}$$

$$L_x=1+\frac{\lambda}{2}(x-1)=0,\ L_y=-1+\frac{\lambda}{2}(y+1)=0,\ L_z=1+2\lambda(z-2)=0$$

$$L_\lambda=\frac{(x-1)^2}{4}+\frac{(y+1)^2}{4}+(z-2)^2-1=0$$

$$\Rightarrow\lambda\neq 0, x=1-\frac{2}{\lambda},\ y=-1+\frac{2}{\lambda},\ z=2-\frac{1}{2\lambda}\Rightarrow\frac{1}{\lambda^2}(1+1+\frac{1}{4})=1$$

$$\Rightarrow\lambda=\pm\frac{3}{2}\Rightarrow f(x,y,z)=4-\frac{9}{2\lambda}$$

$$\Rightarrow f(x,y,z)\text{ 在 }(x,y,z)=(-\frac{1}{3},\frac{1}{3},\frac{5}{3})\text{ 取最小值 }1$$

$$f(x,y,z)\text{ 在 }(x,y,z)=(\frac{7}{3},-\frac{7}{3},\frac{7}{3})\text{ 取最大值 }7$$

19、求旋转抛物面 $z=x^2+y^2$ 与平面 $x+y-2z=2$ 的最短距离.

解: 依题意 $f(x,y,z)=(x+y-2z-2)^2$, 取 $L=f(x,y,z)+\lambda(x^2+y^2-z)$

$$\Rightarrow L_x=2(x+y-2z-2)+2\lambda x=0,\ L_y=2(x+y-2z-2)+2\lambda y=0$$

$$L_z=-4(x+y-2z-2)-\lambda=0,\ L_\lambda=x^2+y^2-z=0$$

$$\Rightarrow 2+2z=(1+\lambda)x+y=x+(1+\lambda)y$$

$$\Rightarrow \lambda=0 \text{ 时 } \begin{cases} x+y-2z-2=0 \\ x^2+y^2=z \end{cases} \Rightarrow 2(x^2+y^2)=x+y-2 \Rightarrow \text{无解}$$

$$\Rightarrow \lambda\neq 0 \Rightarrow \lambda x=\lambda y\Rightarrow x=y,\quad 4\lambda x-\lambda=0\Rightarrow x=y=\frac{1}{4}\Rightarrow z=\frac{1}{8}$$

$$\Rightarrow \lambda=-4(x+y-2z-2)=7$$

$$\Rightarrow d_{\min} \text{ 在 } (x,y,z)=(\frac{1}{4},\frac{1}{4},\frac{1}{8}) \text{ 取得, 取值 } \frac{7}{4\sqrt{6}}$$

7-1、计算 $\iint_D xe^{xy}\,dxdy$, 其中 $D=[0,1]\times[0,1]$

$$(*)=\int_0^1 dx\int_0^1 dy\,xe^{xy}=\int_0^1 dxe^{xy}\Big|_0^1=\int_0^1\left(e^x-1\right)dx=e^x-x\Big|_0^1=e-2$$

7-4、计算 $\iint_D y^2\sqrt{a^2-x^2}\,dxdy$, 其中 $D=\{(x,y)|x^2+y^2\le a^2\}$

解: 由对称性

$$(*)=4\iint_{D_1} y^2\sqrt{a^2-x^2}\,dxdy\quad D_1=\{0\le x\le a,0\le y\le\sqrt{a^2-x^2}\}$$

$$(*)=4\int_0^a dx\int_0^{\sqrt{a^2-x^2}} y^2\sqrt{a^2-x^2}=4\int_0^a dx\sqrt{a^2-x^2}\frac{y^3}{3}\Big|_0^{\sqrt{a^2-x^2}}$$

$$=\frac{4}{3}\int_0^a\left(a^2-x^2\right)^2dx=\frac{4}{3}\int_0^a\left(x^4-2x^2a^2+a^4\right)dx$$

$$=\frac{4}{3}\left(\frac{x^5}{5}-\frac{2}{3}a^2x^2+a^4x\right)\Big|_0^a=\frac{32}{45}a^5$$

7-5、计算 $\iint_D (x^2+y^2)\,dxdy$, 其中 $D: |x|+|y| \le 1$.

解: 由对称性

$$(*) = 4\iint_{D_1} (x^2+y^2)\,dxdy \qquad D_1: 0 \le x \le 1, 0 \le y \le 1-x$$

$$\Rightarrow (*) = 4\int_0^1 dx \int_0^{1-x} dy(x^2+y^2) = 4\int_0^1 dx\left(x^2y + \frac{y^3}{3}\right)\Big|_0^{1-x}$$

$$= 4\int_0^1 dx\left(x^2(1-x) + \frac{(1-x)^3}{3}\right)$$

$$= 4\left(\frac{x^3}{3} - \frac{x^4}{4} - \frac{(x-1)^4}{12}\right)\Big|_0^1 = \frac{2}{3}$$